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Four-Strand Hamstring Tendon Autograft Compared with Patellar Tendon-Bone Autograft for Anterior Cruciate Ligament Reconstruction

A Randomized Study with Two-Year Follow-Up

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ABSTRACT

Seventy-two patients with subacute or chronic rupture of the anterior cruciate ligament were randomly assigned to autograft reconstruction with four-strand gracilis and semitendinosus tendon ($N = 37$) or with patellar tendon-bone ($N = 35$) from the ipsilateral side. The groups were similar in terms of age, sex, level of activity, degree of laxity, meniscal lesions found surgically, and rehabilitation program. The follow-up was performed at another hospital by independent observers after 6, 12, and 24 months. Sixty-one patients (32 with hamstring tendon grafts and 29 with patellar tendon grafts) complied with the follow-up routine for the full 24 months. No differences were found between the groups with respect to Cincinnati functional score, KT-1000 arthrometer measurements, or stairs hopple test results. The subjective result and the single-legged hop test result were better for the hamstring tendon group after 6 and 12 months, but no differences were found after 24 months. The hamstring tendon group showed better isokinetic knee extension strength than did the patellar tendon group after 6 months, but not after 12 and 24 months. There was a significant weakness in isokinetic knee flexion strength among the hamstring tendon group. Anterior knee pain was not significantly different between the groups, but kneeling pain was significantly less common in the hamstring tendon group after 24 months.

The two most commonly used grafts for the repair of a torn ACL are the central third of the patellar tendon, including its bone insertion sites, and the four-strand hamstring tendon graft made of the gracilis and semitendinosus tendons. The patellar tendon graft has been considered the ideal graft choice. It is accessible, has good structural and fixation properties, a potential for bone-to-bone healing, and a predictable success rate in the restoration of knee stability.[1,7,8,11,16,18] However, various donor-site problems have been reported after harvest of patellar tendon grafts. Anterior knee pain, loss of sensation, patellar fracture, inferior patellar contracture, and loss of extension torque impair knee function in spite of a successful replacement of the ACL.[3,6,14,25,27] Therefore, use of the hamstring tendon graft has increased in popularity because of many reports that suggest its use incurs fewer donor-site complications.[5,13,17,27] The structural strength of a hamstring tendon graft with all four strands equally tensioned at time zero (4590 N) is superior to that of a 10-mm patellar tendon-bone graft (2977 N).[8,12] The trend toward increased popularity of the hamstring tendon graft is also related to the development of fixation techniques better than those previously used compared with those used for fixation of the patellar tendon graft.[26]

However, there are few well-designed, randomized studies comparing these two methods. Marder et al.[16] noted no differences between the two methods with respect to subjective complaints, functional level, or laxity, but their

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hamstring tendon group had weaker peak hamstring muscle torque. Both grafts were secured with a single post-and-washer fixation. The reconstructions were performed on an alternating basis and the patients were not randomly selected. Aglietti et al.[1] also compared the two grafts by alternating graft choice in nonrandomized patients. The patellar tendon grafts were fixed with use of a post and washer on the femoral side and an interference screw on the tibial side, whereas the hamstring tendon grafts were fixed with a post and washer outside the femoral and tibial channels. Patellofemoral crepitation and minor loss of extension were more common among patients in the patellar tendon group. Return to sport was more frequent among the patellar tendon graft patients. There were no differences in the incidence of symptoms or laxity between the groups. Corry et al.,[9] in a recent study, compared 82 ACL reconstructions with use of patellar tendon-bone grafts with a succeeding series of 77 ACL reconstructions with use of hamstring tendon graft. A single-incision technique and fixation with interference screws were used in both groups. The authors found no difference between the groups in terms of ligament stability, range of motion, and general symptoms. Thigh atrophy was significantly less in the hamstring tendon group at 1 year but the difference had disappeared by 2 years. Kneeling pain was less common in the hamstring tendon group, suggesting less donor site morbidity.

O'Neill[20] performed a well-designed, randomized study comparing three techniques for ACL reconstruction. One hundred twenty-seven patients with a torn ACL were randomized to receive single-incision reconstruction using patellar tendon graft, a two-incision reconstruction using patellar tendon graft, or a two-incision reconstruction using a double-stranded hamstring tendon graft. The patellar tendon grafts were fixed with interference screws, and the hamstring tendon grafts were fixed with two barbed staples outside the tibial and femoral bone channels. The tendons were left attached to the pes anserinus insertion. The patients treated with a two-incision reconstruction and a patellar tendon graft returned to a greater level of athletic activity than the other two groups, and a higher percentage of the patients in this group had satisfactory stability. However, there were no differences in the overall outcome among the three groups.

Although these studies give good insight into the clinical outcome from use of the two types of grafts, there is still a need for randomized studies that compare other types of fixation and use independent observers and more valid evaluation protocols. The null hypothesis of the present study was that there is no difference in outcome with regard to stability, strength, function, subjective assessment, and donor-site morbidity between ACL reconstructions with hamstring tendon grafts and those with patellar tendon-bone grafts.

MATERIALS AND METHODS

Seventy-two patients with an average age of 26 years (range, 15 to 50) who had a subacute ($N = 18$) or chronic ($N = 54$) rupture of the ACL were randomized to ACL reconstruction with a hamstring tendon autograft ($N = 37$) or a patellar tendon-bone autograft ($N = 35$) from the ipsilateral side. "Subacute" ruptures were those cases in which the patient selected reconstruction after the injury, as soon as the inflammatory reaction was over and full range of motion was restored. "Chronic" ruptures were those cases in which nonoperative treatment was selected by the patient after the injury but the treatment had failed. All of the patients had a positive Lachman and pivot shift test result. Patients with additional ligament injuries before the operation were excluded, but patients who were successfully treated nonoperatively before the operation for a partial medial collateral ligament injury were included. The study was approved by the Norwegian Ethics Committee for Medical Research, and the patients gave their written, informed consent to participate in the study. Two patients refused to participate in the study and selected reconstruction with the patellar tendon autograft. There were no differences between the groups concerning sex, age, and activity level before injury (Table 1). Subacute and chronic ACL injuries were equally distributed among the groups. A partial meniscal resection was performed in 27 patients. In four patients a microfracture procedure was performed because of a chondral lesion on the femoral condyle. The additional surgical procedures were equally distributed between the groups.

The operations were performed in an outpatient department using general anesthesia and a bloodless field. Antibiotic prophylaxis was given before tourniquet inflation as a single dose of 2 g cephalothin. Both groups were operated on using a single-incision technique. A systematic arthroscopic examination was performed initially and then additional surgery was performed. The ACL remnants were cleared away, and a lateral notchplasty and a straightening of the roof of the notch were performed when necessary.

Patellar Tendon-Bone Graft Procedure

For the patients randomized to the group undergoing reconstruction with a patellar tendon graft, a 10-mm graft was harvested via a longitudinal incision. The graft included a 25-mm trapezoidal patellar bone block and a

TABLE 1
Characteristics of the Two Study Groups

Variable	Group	
	Hamstring tendon	Patellar tendon
	Mean (SD)	Mean (SD)
Age (years)	27 (9)	25 (7)
Sex		
Women	16	16
Men	21	19
Time from injury to operation (months)	40.5 (41.6)	41.3 (41.0)
Activity level preinjury[a]	1.5 (0.5)	1.4 (0.6)
Activity level preoperatively[a]	3.1 (0.4)	3.0 (0.2)

[a] According to the International Knee Documentation Committee Evaluation Form.

25-mm rectangular tibial bone block. The bone blocks were trimmed to pass through a 9-mm diameter gauge. A guide wire was drilled from the medial side of the tibial tubercle at a 45° angle to the tibial shaft and advanced to the preserved ligament stump in the posterior portion of the ACL footprint with use of a drill guide (Linvatec Corp., Largo, Florida). A 9-mm cannulated drill bit was used to drill the tibial tunnel. With the knee flexed, a femoral aimer with 7-mm offset (Linvatec Corp.) was introduced through the tibial tunnel and positioned at the 11- or 1-o'clock over-the-top position (right or left knee, respectively). A guide wire was drilled through the aimer and advanced 3 to 4 cm into the femur. A 9-mm cannulated drill bit was introduced over the guide wire and drilled to a depth of 25 mm. The graft was passed into the knee using the Paramax system (Linvatec Corp.) and fixed with a 7 × 25 mm titanium femoral interference screw (Linvatec Corp.). The knee was cycled under graft tensioning to allow stress relaxation of the graft. The graft was then tensioned to 20 pounds and, with the knee at 0° of flexion, fixed in the tibial tunnel with a 7 × 25 mm tibial interference screw.

Hamstring Tendon Graft Procedure

For the patients in the group undergoing reconstruction with the hamstring tendon graft, the graft was harvested through a longitudinal incision at the site of the pes anserinus insertion. The sartorius fascia was split, and the gracilis and semitendinosus tendons were harvested with a tendon stripper (Smith & Nephew Endoscopy, Inc., Andover, Massachusetts). The tendons were cleaned of adherent muscle fibers. A graft preparation device (Graft Master, Smith & Nephew) was used to tension the tendons, and the free ends of both tendons were sutured together with No. 2 polyester suture in a running baseball-style whipstitch. The tendons were looped over a 5-mm polyester tape (Endotape, Smith & Nephew) to create a quadruple graft, and the graft was sized between 7 and 9 mm. A titanium button (EndoButton, Smith & Nephew) was placed into the holder on the Graft Master, and the Endotape was passed through its two central holes. The tape loop was provisionally secured by a hemostat. The span of tape was set later, once the length of the femoral tunnel was measured. The graft was pretensioned to 20 pounds on the Graft Master while the remainder of the procedure was completed.

The tibial tunnel was prepared according to the method used for the patellar tendon grafts, but the size of the drill bit was selected according to the graft size. For the femoral tunnel preparation, a 5.5-mm offset femoral aimer was used. The guidewire was advanced completely through the femoral cortex and overdrilled by a 4.5-mm drill bit. A depth gauge was used to measure the length of this tunnel. A closed-end femoral socket was drilled 25 mm into the femur with an additional 10 mm for the EndoButton to clear the femoral cortex and flip. The length of the tunnel was used to set the EndoButton tape length, and the tape was tied using two square knots. The EndoButton with the attached graft was pulled by sutures into the joint and flipped as it passed through the femoral tunnel to fix the graft. The knee was cycled under graft tensioning to settle the EndoButton and to allow stress relaxation of the graft. The graft was tensioned with 20 pounds distally and fixed with an interference screw (RCI, Smith & Nephew) in the tibial tunnel with the knee extended. Additionally, a backup fixation of the tendon ends was performed using two barbed staples outside the tibial tunnel.

Rehabilitation

After skin closure and wound dressing, a cold compression device (Aircast, Inc., Summit, New Jersey) was applied and used during the 1st postoperative week. Both groups went through the same aggressive rehabilitation program,[28] which did not include use of a brace or continuous passive motion. Weightbearing was permitted as tolerated from day 1. After 2 weeks, closed kinetic chain exercises and stationary bicycling were started. After 6 weeks, agility training and light jogging were permitted. After 10 weeks, increased agility workouts, maximum strength training, and sport-specific activities were permitted. The patients were permitted to go back to full sports participation after 6 months if the criteria of full range of motion, no effusion, and good knee stability and strength were met.

Follow-up Evaluation

Follow-up evaluation was performed at 6, 12, and 24 months after surgery by independent observers (IH and MAR) in another hospital. Knee joint laxity (anterior displacement of the tibia relative to the femur) was recorded using a KT-1000 knee arthrometer (MEDmetric Corp., San Diego, California) at the manual maximum force.[29] The difference in displacements between the injured and uninjured knee was recorded. Knee function was evaluated with the Cincinnati knee score system,[19] which has a maximum score of 100 points. The Cincinnati questionnaire consists of questions regarding symptoms and function: pain, swelling, giving way, general activity level, walking, stair climbing, running, jumping, and twisting activities. Anterior knee pain was defined as less than the maximum pain-free score of 20 points in the pain variable of the Cincinnati score when the pain was located in the anterior part of the knee. Isokinetic strength was measured with the Cybex 6000 dynamometer (Cybex International, Inc., Medway, Massachusetts). Before testing, the patients performed aerobic ergometer cycling for 8 minutes. The test protocol consisted of 5 repetitions at an angular velocity of 60 deg/sec, followed by a 1-minute rest period and then 30 repetitions at 240 deg/sec. Isokinetic parameters used for evaluation of muscle function were total work at 60 deg/sec (strength) and total work at 240 deg/sec (endurance). Two functional tests (the stairs hopple test and the single-legged hop test) were used to evaluate lower limb function.[22] The stairs hopple test was performed by the patients jumping up and down 22 steps (each step was 17.5 cm in height) on a staircase, followed by the corresponding procedure with the involved leg. The

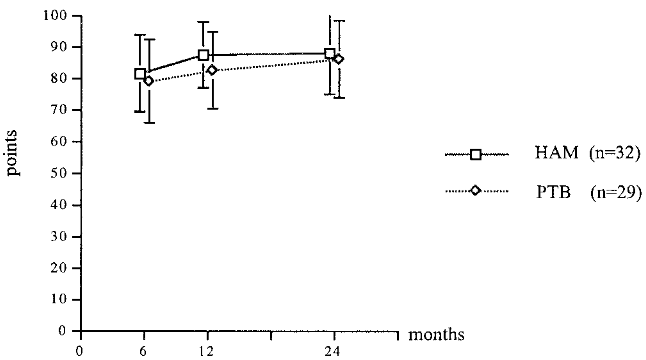

Figure 1. Knee function evaluated with the Cincinnati knee scoring system after reconstruction of the ACL with a four-strand hamstring tendon autograft (HAM) or a patellar tendon-bone autograft (PTB). Values are means ± 1 SD.

time was measured in seconds, and the difference between the uninvolved and involved leg was recorded. A visual analog scale was used to record patient satisfaction at each follow-up interval and kneeling pain after 24 months.[14,21] The kneeling test is a new test in which patients are asked to kneel and to walk on their knees to assess the degree of discomfort in the involved knee compared with the uninvolved knee.[14]

Numbers are presented as arithmetic means and dispersion by ±1 standard deviation. The results were analyzed according to the method of intention to treat. The differences between the groups were analyzed with use of repeated-measures analysis of variance. Statistical significance was set at $P < 0.05$.

RESULTS

Three patients were excluded because of ACL injury of the primary uninvolved knee during the study period. Eight patients were lost to follow-up. This left 61 patients who complied with the follow-up routine after 24 months; 32 were in the hamstring tendon group and 29 in the patellar tendon group.

Two patients in the hamstring tendon group suffered postoperative complications. In one patient, a lesion of the saphenous nerve caused a permanent loss of sensation on the medial aspect of the tibia. In another patient, a rupture of the sartorius tendon caused a severe flexion-strength deficit that was verified by MRI 6 months after the operation. These two complications were related to graft harvesting. Five patients underwent a second operation during the study period. In three patients, two from the hamstring tendon group and one from the patellar tendon group, an ACL revision was performed because of traumatic graft failure. In two patients, both from the patellar tendon group, a second arthroscopic procedure was performed. In one of the cases, a meniscal rupture was addressed, and in the other a cyclops lesion was resected. Loss of range of motion compared with the control knee was not present in any patient in either group.

No differences were found between the groups with respect to Cincinnati functional score (Fig. 1), KT-1000 arthrometer measurements, or stairs hopple test results. Patient satisfaction and the result of the single-legged hop test were superior in the hamstring tendon group after 6 and 12 months, but no differences were found between groups after 24 months. Kneeling pain was significantly less common in the hamstring tendon group after 24 months (Table 2). Although anterior knee pain was slightly more common in the patellar tendon group, no statistically significant difference was found (Table 2). Isokinetic testing at 60 and 240 deg/sec showed that the hamstring tendon group had better isokinetic knee extension strength and endurance after 6 months compared with the patellar tendon group. After 12 and 24 months, no differences were found (Figs. 2 and 3). There was a significant weakness in isokinetic flexion strength at 60 deg/sec in the hamstring tendon group compared with the patellar tendon group at 12 months. At 240 deg/sec, the endurance flexion weakness was significant at all follow-up intervals (Figs. 4 and 5).

DISCUSSION

The aim of this study was to compare patellar tendon and hamstring tendon autografts in a randomized study that

TABLE 2
Comparison of Outcomes in the Hamstring Tendon Group (HAM) and the Patellar Tendon Group (PTB)

Outcome	6 months		12 months		2 years	
	HAM Mean (SD)	PTB Mean (SD)	HAM Mean (SD)	PTB Mean (SD)	HAM Mean (SD)	PTB Mean (SD)
KT-1000 arthrometer manual maximum difference (mm)	2.9 (2.4)	3.3 (2.6)	2.8 (2.5)	2.6 (2.2)	2.7 (2.1)	2.7 (2.2)
Patient satisfaction (VAS)[a]	87.8 (16.3)[b]	76.3 (25.4)	87.6 (14.1)[b]	77.8 (22.6)	84.8 (27.5)	78.4 (26.4)
Cincinnati score	81.4 (12.2)	79.0 (13.2)	87.1 (10.5)	82.4 (12.3)	87.8 (13.0)	85.9 (12.2)
Anterior knee pain (incidence)	5.3%	14.4%	8.8%	10.5%	12.5%	16.1%
Kneeling problems (VAS)					18.9 (26.2)	35.5 (33.6)[b]
Stairs hopple test (% loss compared with uninvolved)	8.7 (11.4)	17.2 (29.7)	10.5 (18.3)	12.6 (18.5)	2.6 (10.2)	6.5 (13.7)
Single-legged hop test (% loss compared with uninvolved)	3.9 (4.8)[b]	12.3 (11.9)	3.1 (6.7)[b]	7.9 (7.8)	0.7 (6.9)	4.3 (7.3)

[a] VAS, visual analog scale.
[b] Significant difference between groups ($P < 0.05$).

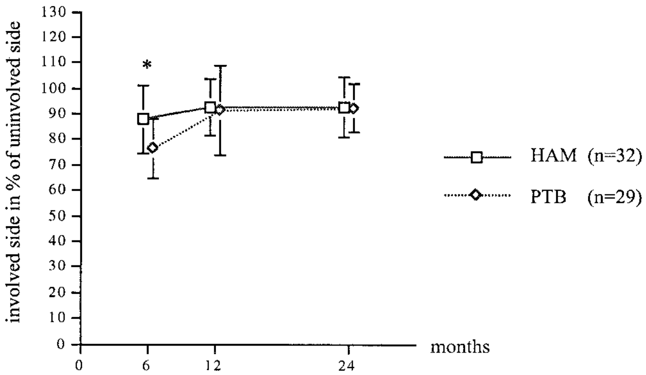

Figure 2. Relative extension at the angular velocity of 60 deg/sec after reconstruction of the ACL with a four-strand hamstring tendon autograft (HAM) or a patellar tendon-bone autograft (PTB) during the 2-year follow-up period. Values are means ± 1 SD. The asterisk indicates a significant difference at 6 months ($P < 0.001$).

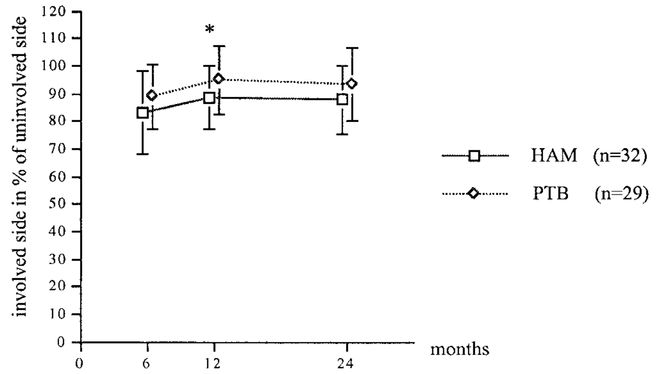

Figure 4. Relative flexion at the angular velocity of 60 deg/sec after reconstruction of the ACL with a four-strand hamstring tendon autograft (HAM) or a patellar tendon-bone autograft (PTB) during the 2-year follow-up period. Values are means ± 1 SD. The asterisk indicates a significant difference at 12 months ($P < 0.05$).

used independent observers to assess outcomes. The results in terms of patient satisfaction, functional ability, quadriceps muscle strength, and endurance were better in the hamstring tendon group at 12 months after the operation. After 2 years, however, these parameters had equalized. The hamstring tendon group had less anterior knee pain when kneeling, but there was a lack of hamstring muscle strength and endurance compared with the patellar tendon group, showing a difference in donor-site morbidity between the groups.

There are several limitations to this study. Although susceptibility bias was limited by the randomized study design, there is still a possible performance bias. Although one surgeon (AKA) treated both groups and the same rehabilitation protocol was used for both, two different fixation techniques were used for the two grafts. Therefore, this study compared not only the use of two different grafts but also two different techniques of fixation. Corry et al.,[9] in a recent study, used the same interference screw fixation technique for both hamstring tendon and patellar tendon grafts, with no difference between the groups in terms of ligament stability. We did not use this type of fixation in the present study because of reports of inferior pull-out strength and stiffness of interference screw fixation of hamstring tendon grafts in vitro.[2]

The failure load and stiffness for the EndoButton construct used in this study (692 ± 48 N and 132 ± 64 N/mm, respectively) matches the data for interference screw fixation of patellar tendon grafts.[26] Recently, an EndoButton (EndoButton CL) has been developed with a factory-attached continuous polyester loop, eliminating the need for knot tying of the connector material. This change in design has increased the failure load and stiffness to 1430 ± 115 N and 155 ± 24 N/mm, respectively.[4] The EndoButton technique has been criticized because of the greater distance of this suspensory type

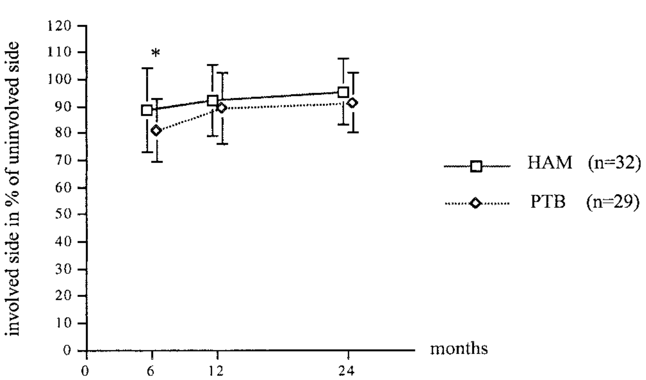

Figure 3. Relative extension at the angular velocity of 240 deg/sec after reconstruction of the ACL with a four-strand hamstring tendon autograft (HAM) or a patellar tendon-bone autograft (PTB) during the 2-year follow-up period. Values are means ± 1 SD. The asterisk indicates a significant difference at 6 months ($P < 0.05$).

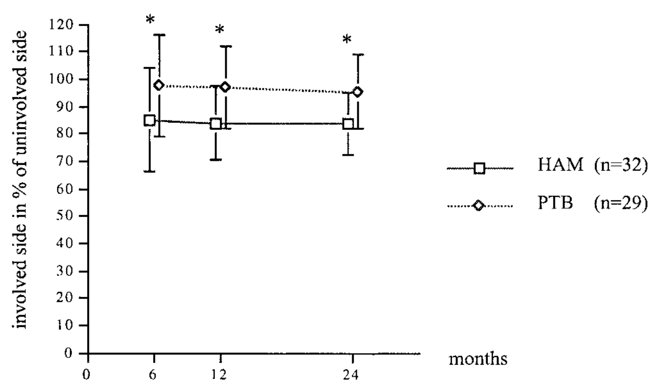

Figure 5. Relative flexion at the angular velocity of 240 deg/sec after reconstruction of the ACL with a four-strand hamstring tendon autograft (HAM) or a patellar tendon-bone autograft (PTB) during the 2-year follow-up period. Values are means ± 1 SD. The asterisks indicate a significant difference between groups ($P < 0.01$).

of fixation than fixation at the tunnel apertures with interference screws or cross pins. It has been hypothesized that the EndoButton construct will exhibit substantial graft-tunnel motion because of stretch in the polyester tape. The literature is not consistent on this point. In some studies, cadaveric tests under cyclic loading have shown substantial graft-tunnel motion of 3 to 5 mm (F. Fu et al., unpublished data, 1998), whereas others have detected less than 1 mm motion, corresponding to that of a patellar tendon graft fixed with an interference screw.[4] On the tibial side, interference fixation was used for both groups, with an additional backup of staples for the hamstring tendon grafts.

Tibial placement was similar for the two grafts, and in both techniques the grafts were placed in a near over-the-top position on the femoral side. Tensioning at fixation was similar; however, we used the Graft Master to pretension the hamstring tendon grafts but not the patellar tendon grafts. The KT-1000 arthrometer data showed similar side-to-side differences, suggesting no difference because of type of graft, although there was some methodologic bias in tensioning and fixation of the grafts. These results correspond to those of previous reports in which no difference was detected between hamstring tendon and patellar tendon grafts in terms of ligament stability, although different types of fixation for the hamstring tendon grafts were used.[1,9,16,20]

Detection bias was limited by use of independent observers in another hospital and outcome assessments that revealed satisfactory reliability and a good sensitivity to clinical changes over time.[23,24] The exclusion of patients and loss to follow-up indicates a transfer bias. Patients treated with ACL reconstruction are not easy to follow up because they may move out of the city or abroad because of sports events and educational pursuits. The uninvolved knee was used as a control in this study. Three patients suffered an ACL rupture of the uninvolved knee during the study period and had to be omitted from further follow-up. However, the patients lost to follow-up were equally distributed between the groups, and 85% is a high rate of follow-up in this type of study.

The study population was small, and the size may not have allowed for detection of small differences between groups. Some data, such as the results of functional tests, were nearly significant at 12 and 24 months in favor of the hamstring tendon group, suggesting a type II error on these parameters. But knee joint laxity was used as the primary outcome measure to determine the number of patients required in each group. The study was planned to detect major differences in knee joint laxity of 3 mm between groups. The standard deviation for the KT-1000 arthrometer side-to-side difference was set at 3 mm or more. With alpha set to 0.05 (type I error) and beta to 0.01 (type II error), a power analysis demonstrated that a minimum of 22 patients was required in each group to complete the trial. Therefore, 72 patients were enrolled to allow for the high risk of dropouts in this ACL study population.

Quadriceps muscle weakness is often encountered when the patellar tendon is used for ACL reconstruction.[25] Marder et al.[16] and Aglietti et al.[1] did not report any differences in isokinetic performance in knee extension between patients operated on with a patellar tendon graft or a hamstring tendon graft, in contrast to the results of the present study. Some quadriceps muscle weakness may be related to the ACL injury and the operation, rather than to the source of the graft. Although both groups went through an accelerated rehabilitation program, the isokinetic total work in knee extension both at 60 and 240 deg/sec was significantly reduced in the patellar tendon group compared with the hamstring tendon group after 6 months. This result suggests that the quadriceps muscle weakness is partly related to patellar tendon donor-site morbidity. The reduced performance during the single-legged hop test and the reduced satisfaction of the patellar tendon group patients seemed to be related to the loss of knee extension strength because, among impairments and disabilities, quadriceps muscle performance is a main factor related to patients' perception of knee function.[25]

Isokinetic hamstring muscle strength and endurance test results showed significant differences in favor of the patellar tendon group at all test intervals. Weakness of the hamstring muscles was also reported by Marder et al.[16] In other reports, however, reduced hamstring muscle strength has not been a major problem when the hamstring tendons are used to provide graft material.[1,10,20,30] Lipscomb et al.[15] reported that mean hamstring muscle strength in the ACL-reconstructed leg was 99% of that of the contralateral leg after reconstruction with an autologous hamstring tendon graft. Furthermore, regeneration of the gracilis and semitendinosus tendons after harvest for ACL reconstruction has been demonstrated.[10]

In the present study, both groups participated in the same rehabilitation program, and no additional training of the hamstring muscles was provided for the patients whose knees were reconstructed with hamstring tendon graft. The hamstring muscle strength deficit may have been secondary to inadequate rehabilitation of the hamstring tendon group. Marder et al.[16] related their findings to the fact that they did not repair the incision in the sartorius fascia, did not perform isokinetic training in the postoperative period, and because they included patients whose sports activity was strictly recreational and who thus may not have been motivated to achieve maximal rehabilitation. In the present study, the sartorius fascia was closed, but there was still a hamstring muscle strength deficit. We propose that patients who receive a hamstring tendon graft need to follow an aggressive strengthening program for the knee flexors as part of their training program during rehabilitation, even after they have returned to normal sports activities.

One of the donor-site problems encountered after harvest of a patellar tendon graft is persistent anterior knee pain.[27] The relationship of the graft type to anterior knee pain is unclear. Marder et al.[16] noted that 24% of their patients reported anterior knee pain, but no difference with respect to the type of graft was observed. Aglietti et al.[1] did not find any significant differences between the groups with respect to patellofemoral symptoms, although patellofemoral crepitus was more frequent in the patellar tendon group. O'Neill[20] reported infrequency of anterior knee pain with either technique and estimated that this was a result of an aggressive rehabilitation program that eliminated flexion contractures. We found no difference between our groups with regard to anterior knee pain. Previous studies have not used reliable tests to record anterior knee pain. Kartus et al.,[14] in a recent study, reported that 51% of their patients had severe problems or were unable to kneel or walk on their knees after ACL reconstruction with a patellar tendon graft. These problems were correlated to loss of hyperextension or sensitivity in the operated area. The inability to kneel and walk on one's knees can be very disturbing for patients in certain occupations and sports, and in many activities of daily life. There was significantly less kneeling pain among the hamstring tendon group than among the patellar tendon group in the present study. The present study is the first comparative study that uses the kneeling test to detect anterior knee pain. The kneeling test is a functional and reliable test for evaluating donor-site morbidity.[14]

CONCLUSIONS

The results of the present study showed a trend toward better subjective results, quadriceps muscle strength, endurance, and functional performance after 6 months when a hamstring tendon graft had been used rather than a patellar tendon graft, but these parameters equalized with time. Significant hamstring muscle weakness in the patients with hamstring tendon grafts persisted until 2 years after the operation, but less donor-site pain was present after harvest of the hamstring tendon graft compared with the patellar tendon graft.

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